FFT & Signal Playground

Formula sheet

Every quantity below is the one computed by `signals.py`. Symbols: f_s sample rate, N number of samples, $x[n]$ signal, $w[n]$ window, k frequency bin.

1 Sampling

$$T = \frac{N}{f_s} \qquad \Delta f = \frac{f_s}{N} \qquad f_{\text{Nyq}} = \frac{f_s}{2} \qquad f_k = k\,\Delta f \quad (k = 0, \dots, N/2)$$

$$f_{\text{alias}} = \left| f - f_s \,\text{round}(f/f_s) \right| \qquad \text{(a tone with } f > f_{\text{Nyq}} \text{ appears at } f_{\text{alias}})$$

Resolving two tones needs a separation of at least about the main-lobe width of the window: $|f_2 - f_1| \gtrsim 2\Delta f$ for a Hann window. Finer resolution needs a longer record (larger N), not a higher f_s.

2 Discrete Fourier transform

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N} \qquad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]\, e^{j2\pi kn/N}$$

$$\sum_n |x[n]|^2 = \frac{1}{N} \sum_k |X[k]|^2 \quad \text{(Parseval)} \qquad X[N-k] = X[k]^* \text{ for real } x$$

The real-input FFT keeps only $k = 0, \dots, N/2$. An FFT costs $O(N \log N)$ operations against $O(N^2)$ for the direct sum.

3 Windowing and amplitude scaling

The spectrum of the windowed signal is the true spectrum convolved with the window's transform. Define the *coherent gain*

$$\text{CG} = \frac{1}{N} \sum_n w[n] \qquad \Longrightarrow \qquad A_k = \frac{2\,|X_w[k]|}{N \cdot \text{CG}} \quad (1 \le k < N/2)$$

so a sine of amplitude A on a bin centre reads exactly A for every window. The DC and Nyquist bins are not doubled. Levels in the plots are $20 \log_{10} A_k$ (dB re amplitude 1). The equivalent noise bandwidth is

$$\text{ENBW} = N \, \frac{\sum_n w[n]^2}{\left(\sum_n w[n]\right)^2} \text{ bins}$$

Window	$w[n]$ for $0 \le n < N$	CG	ENBW	Null at	Peak sidelobe	Worst scallop
Rectangular	1	1.000	1.00	± 1 bin	-13.3 dB	-3.90 dB
Hann	$0.5 - 0.5\cos\frac{2\pi n}{N}$	0.500	1.50	± 2 bins	-31.5 dB	-1.42 dB
Hamming	$0.54 - 0.46\cos\frac{2\pi n}{N}$	0.540	1.36	± 2 bins	-42.7 dB	-1.75 dB
Blackman	$0.42 - 0.5\cos\frac{2\pi n}{N} + 0.08\cos\frac{4\pi n}{N}$	0.420	1.73	± 3 bins	-58.1 dB	-1.10 dB

Scalloping is the amplitude error when a tone falls halfway between two bins (last column). A narrower main lobe (rectangular) gives the best resolution but the worst leakage. A wider lobe (Blackman) hides weak tones next to strong ones.

4 Peak frequency by parabolic interpolation

With the largest bin k and its neighbours $\alpha = A_{k-1}$ and $\beta = A_k$ and $\gamma = A_{k+1}$:

$$\delta = \frac{1}{2}\,\frac{\alpha - \gamma}{\alpha - 2\beta + \gamma} \qquad f_{\text{peak}} = (k + \delta)\,\Delta f \qquad A_{\text{peak}} \approx \beta - \tfrac{1}{4}(\alpha - \gamma)\,\delta \qquad (-\tfrac{1}{2} \le \delta \le \tfrac{1}{2})$$

This locates a tone to a small fraction of a bin (about 0.1 Hz in the app at $\Delta f \approx 2$ Hz).

5 Filters

Magnitude responses used in the app (n is the filter order f_c the cutoff):

$$|H_{\mathrm{LP}}(f)| = \frac{1}{\sqrt{1+(f/f_c)^{2n}}} \qquad |H_{\mathrm{HP}}(f)| = \frac{1}{\sqrt{1+(f_c/f)^{2n}}}$$

$$|H_{\mathrm{BP}}(f)| = \frac{1}{\sqrt{1+u^{2n}}} \qquad |H_{\mathrm{BS}}(f)| = \frac{1}{\sqrt{1+u^{-2n}}} \qquad u = \frac{f^2 - f_0^2}{f\,B} \quad f_0 = \sqrt{f_1 f_2} \quad B = f_2 - f_1$$

- $|H| = 1/\sqrt{2}$ (-3 dB) at every cutoff. The roll-off is $20n$ dB per decade ($6n$ dB per octave).
- The filter is applied as $Y[k] = H(f_k)\,X[k]$ on a copy zero-padded to the next power of two $\geq 2N$, then cropped back to N samples. Because H is real the output has **zero phase shift**, and the padding keeps circular wrap-around out of the result.
- Convolution in time is multiplication in frequency: $y = x * h \iff Y = X\,H$.

6 Spectrogram (short-time Fourier transform)

$$S[m,k] = \sum_{n=0}^{L-1} x[n+mH]\,w[n]\,e^{-j2\pi kn/L} \qquad \mathrm{level}[m,k] = 20\log_{10}\frac{|S[m,k]|}{\sum_n w[n]/2}$$

with a Hann window of length $L \leq 256$ and hop $H = L/4$. Frequency resolution is f_s/L and time resolution is about L/f_s, so their product is fixed: a longer window sharpens frequency and blurs time. Values are clipped at -80 dB.

7 Useful theorems and pairs

Property	Statement	Pair	Transform
Linearity	$ax + by \leftrightarrow aX + bY$	Rectangle (width T)	$T\,\mathrm{sinc}(fT)$
Time shift	$x[n-n_0] \leftrightarrow X[k]\,e^{-j2\pi kn_0/N}$	Gaussian	Gaussian
Modulation	$x[n]\,e^{j2\pi k_0 n/N} \leftrightarrow X[k-k_0]$	Impulse train	Impulse train
Multiplication	$xy \leftrightarrow X * Y\,/\,N$	Pure sine	two spectral lines

The *Morse code* preset is a keyed carrier $m(t)\sin(2\pi f_0 t)$. By the modulation theorem its spectrum is the spectrum of the on/off envelope $m(t)$ shifted to $\pm f_0$. Faster keying makes $m(t)$ change faster, so the sidebands spread wider around the carrier. The 5 ms raised-cosine edges used in the app make the sidebands fall off much faster than hard on/off keying, which is what removes key clicks.

8 Worked example

For $f_s = 8000$ Hz and $N = 4096$:

$$T = 0.512\ \mathrm{s} \qquad \Delta f = 1.953\ \mathrm{Hz} \qquad f_{\mathrm{Nyq}} = 4000\ \mathrm{Hz} \qquad \text{a 5 kHz tone aliases to } |5000 - 8000| = 3000\ \mathrm{Hz}$$

A 4th-order low-pass at $f_c = 900$ Hz passes a 500 Hz tone with $|H| = \left(1 + (500/900)^8\right)^{-1/2} \approx 0.995$ and a 1300 Hz tone with $|H| = \left(1 + (1300/900)^8\right)^{-1/2} \approx 0.224$.